

Allen Dufort

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EDUCATION

Brown University

Master of Science in Computer Science, GPA: 4.0

Bachelor of Science in Computer Science, GPA: 3.85

Providence, RI

May 2025

May 2024

TECHNICAL SKILLS

Languages: Python, SQL, Golang, JavaScript, Java, HTML, CSS, C

Entry-level Knowledge of Frameworks: Terraform, Apache Spark, MongoDB, AWS, Azure, WordPress

Areas of Expertise: Generative AI, Large Language Models, AI Programming, AI Ethics, Data Science, Predictive Modeling, Machine Learning, Deep Learning, Database Analysis, Web Design, UI/UX, Quality Assurance Testing, Git

EXPERIENCE

Graduate Teaching Assistant, *Brown University*

Providence, RI | **January 2025 – May 2025**

- Helped create content to teach students how to make multi-agent systems using generative AI and create large language models
- Assisted in grading course assignments to provide comprehensive feedback and assessment of student performance

Software Engineering Intern, *Marqeta*

Remote | **June 2024 – August 2024**

- Developed a GitHub application integrated with the CI team's service, issuing on-demand personal access tokens scoped to user-specified repositories
- Reduced token retrieval time by 95 percent, enhancing the CI/CD pipeline's efficiency

Software Engineering Intern, *General Motors*

Roswell, GA | **June 2023 – August 2023**

- Implemented front-end features for Account Overview used by over 22 million online users
- Led testing phase, automating, and executing tests to identify critical bugs and enhance product quality
- Engaged in scrum practices, including reviewing and creating PRs for various tickets

Software Engineering Intern, *Brown University Physics Department*

Providence, RI | **November 2021 – May 2024**

- Took ownership of the cryostat control system's backend of the Exoplanet Climate Infrared Telescope (EXCITE)
- Collaborated on backend code planning with NASA's Goddard Space Flight Center and other institutions

Freelance Web Developer, *Diabetic and Hypertension Relief Center*

Remote | **January 2015 – Present**

- Oversee UI design, including CSS/HTML elements, text updates, and photo uploads, for the WordPress-supported site dhreliefcenter.org since 2021

PROJECTS

Language Models as Reward Functions for RL | *Python, Jupyter Notebook, Gymnasium, Hugging Face*

Spring 2025

- Developed reinforcement learning framework using large language models (LLMs) as reward signal generators instead of traditional mathematical reward functions
- Implemented and evaluated Q-learning agents in Frozen Lake and Blackjack environments, comparing LLM-based rewards against numeric baselines across 400–500 training episodes
- Demonstrated that language-driven rewards can provide more flexible, intuitive, and human-aligned feedback for training reinforcement learning agents

Iterative Design | *HTML, CSS, Figma, Balsamiq*

Spring 2025

- Redesigned Sesame's healthcare platform, enhancing user accessibility and personalization for AI-driven healthcare tools
- Developed iterative prototypes (Balsamiq, Figma) based on expert feedback from Google and Sesame executives, refining UX through structured critique cycles

JeopardyAI! | *Python, Jupyter Notebook, AutoGen, OpenAI*

Fall 2024

- Developed Jeopardy-style trivia game where human players can play against each other and interact directly with AI game host
- Designed multi-agent system that dynamically generated questions, moderated gameplay, and kept score
- Integrated OpenAI and Cluebase APIs to deliver an engaging experience with access to over 350,000 trivia questions.

HONORS AND AWARDS

Brown University Computer Science Scholarship

March 2024

- Merit-based full-tuition scholarship for the Master's Program of Computer Science at Brown

Brown University Undergraduate Teaching and Research Award

Spring 2024, Spring 2023, Fall 2022

- Supports Brown students collaborating with Brown faculty on research

NASA Rhode Island Space Grant Summer Scholarship

April 2022

- Awarded to undergraduates carrying out NASA-related research projects

Dr. Martin Luther King Jr. Trust Fund Scholarship

October 2019

- Awarded to college-bound students with exceptional academics and community service experience